

03

Failure Mechanisms

Comprehensive Study Notes

Course Instructor

Dr. Hanadi Salem

[linkedin.com/in/hanadi-salem-11361a11](https://www.linkedin.com/in/hanadi-salem-11361a11)

Compiled by

Abdelrahman Sawy

[linkedin.com/in/abdelrahman-sawy-542740295](https://www.linkedin.com/in/abdelrahman-sawy-542740295)

The American University in Cairo (AUC) — School of Sciences and Engineering

Chapter 3: Failure Mechanisms

Failure Mechanisms

Ductile Materials

[Fail on the planes of maximum shear which means planes tend to slide relative to other planes. When planes cannot move any more, they shear along those planes. Movement of dislocations → plastic deformation]

Elastic Failure

If the elastic modulus E is too low: the component may deflect too much in service, making it unusable. Examples are moving parts such as engines and structures such as bridges.

Alternatively, some components may not deflect as much as they should do or may require too much force to deflect by the required amount. Examples are switches and springs. Such failures become apparent instantly as soon as the load is applied, except in viscoelastic materials such as polymers, where elastic strain can build up gradually over time. Elastic deflection can be distinguished from plastic strain and creep strain because it is totally recoverable on removal of the load.

Buckling:

Though buckling leads to a failure by another mechanism (usually ductile failure), the process of buckling is essentially an elastic instability, controlled by the material's elastic modulus. A classic example is the failure of slender columns loaded in compression. This type of failure generally occurs instantaneously; Difficult to predict because it depends on the shape of the part and how it is loaded and restrained.

A failure is termed "ductile" if it occurs by plastic deformation, i.e., if the yield strength of the material is exceeded. This causes a permanent change in shape of the part, which becomes apparent if one attempts to reassemble broken components. Tensile stress causes necking, reducing the cross section of the part.

Ductile materials fail on planes of maximum shear ($45^\circ \rightarrow$ shear band). Crack propagation will only occur along the shear bands which lie on the planes of maximum shear. Brittle materials fail on plains of maximum tension. Cracks propagation ordinary behavior is to take the shortest path. Crack propagation blunts when faced by empty spaces.

Crack propagation in ductile materials is either transversely (trans granular failure: cracks cuts through the grain boundaries and grains – materials ordinary behavior: properties along the boundaries are the same as within the grain) or intergranular failure (crack propagates along the grain boundaries). Common feature between than $\rightarrow$ rough dirty grey surface. Ductile materials with uniform mechanical properties and uniform particles distribution will fail trans granular. Defects in the grain boundaries will lead to intergranular failure. This may happen due to:

- 1) brittle particles present within the grain boundaries or
- 2) particles are preferentially precipitated on the grain boundaries forming voids or
- 3) brittle phase present on the boundaries (due to sulfurization, incipient melting, or sensitization creating a ductile material with brittle grain boundaries) $\rightarrow$ Brittle Intergranular Failure. In ductile trans granular failure with stresses that are much higher than blunting (σ crack tip $\gg \gg \sigma$ yield), crack propagates by creating a plastic zone and spreading until reaching a particle which creates multiple voids.

Ductile failure occurs by (Void Sheet Mechanism):

- 1) Void Nucleation
- 2) Void Growth
- 3) Void Coalescence

a) Cup and Cone Failure (Tensile – Uniaxial Loading)

Stresses are uniformly distributed along the sample cross-section and length leading to uniform elongation and cross-section reduction. After exceeding the ultimate tensile strength, necking starts due to opening of voids or cracks. At the necked region, the state of stress that is developed triaxial.

Failure Mechanism: Void Sheet (Ductile Failure Mechanism)

Plastic zone → no motion of dislocations.

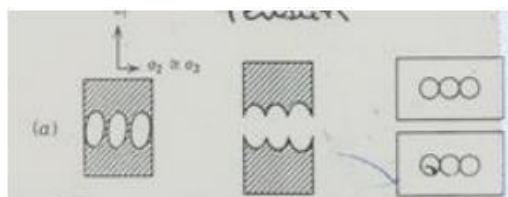
Yield point → beginning of motion of dislocations.

Between yield and ultimate → multiplication of dislocations and stopping motion.

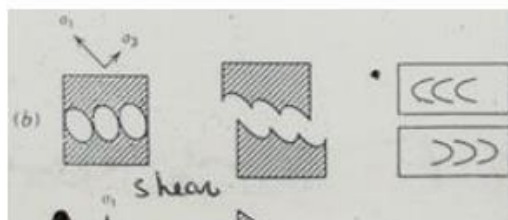
σ Ultimate → dislocations are stopped creating stress concentrations.

Mechanism:

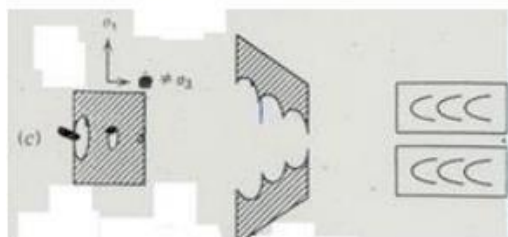
1. Void growth in a triaxial stress state.
2. Crack and deformation bands formation.
3. Nucleation of smaller particles along the deformation bands.
4. Cup and cone fracture in uniaxial tension.



a) Under simple uniaxial stress



b) Under shear stress (elongated dimples along plans of maximum shear)



c) Under tension and bending stress Dimples point back towards the crack origin

1. Investigation of the fracture surfaces at low magnifications will reveal: Rough, spongy texture.
2. At high magnification using SEM will show dimples
 - Dimples are an ordered array of couplets that are found on the surface. Each of these couplets are oriented in a single direction depending on the orientation of the applied stresses. Elongation of the pores parallel to the tensile axis occurs at the onset of the void sheet separation (split). Under shear stress (elongated dimples along plans of maximum shear) that depends on the state of stress.

b) **Mechanism of Ductile Failure**

1. Void nucleation:
 - Inclusion/particle/precipitate size
 - Level of stress
 - Strain rate level
 - Local deformation mode (depends on difference in strength and hardness of the inclusion relative to matrix surrounding it)
 - Alloy purity
2. Void Growth:
 - Occurs by plastic flow of the matrix surrounding the inclusion which promotes decohesion of the small particles from the matrix.
3. Void Coalescence:
 - Is the joining and coalescence of the countless micro voids into a large crack.

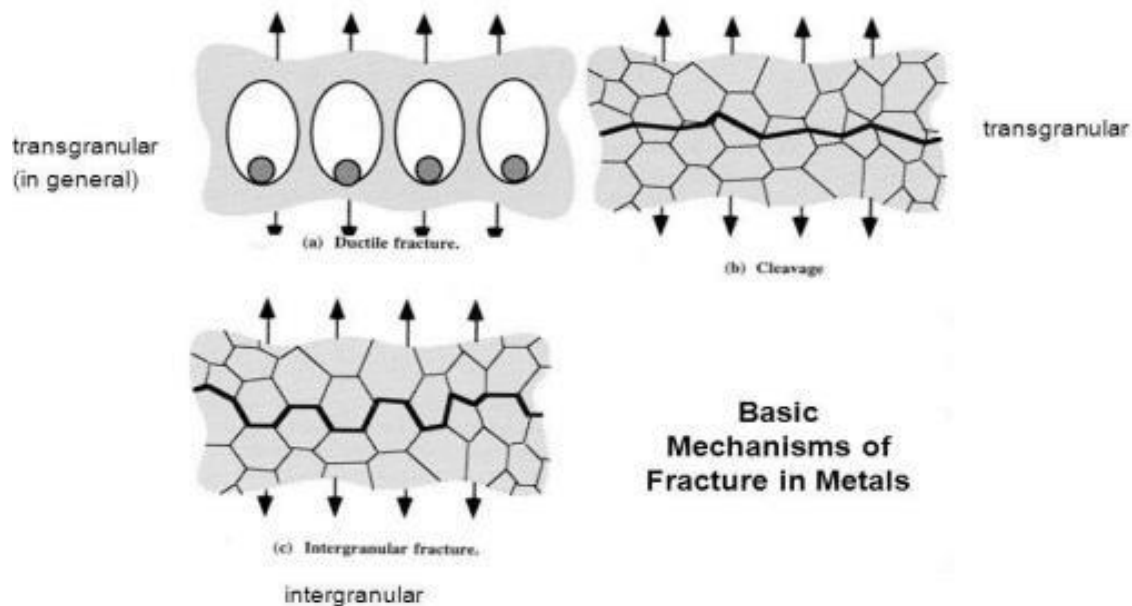
Factors promoting this mechanism:

- Void nucleation can also occur due to particle cracking.
- Interfacial failure between an inclusion or a ppt./particle and the surrounding matrix
- When the spacing between adjacent inclusions, particles or ppt. is too small.

Mechanism of Fracture in Al-Li-Mg Alloys:

1. Slip bands on particles.
2. Intruding slip bands fracture the second phase particles.
3. Void growth under the action of hydrostatic pressure
4. Decohesion occurs at the end of cracked particles.
5. Void coalesces and propagates down grain boundaries.

c) Types of Ductile Failures



d) Macro-Fracture Features

Rough Dirty Grey Surface/Spongy micro feature → Ductile Failure (Maybe Ductile-Brittle Failure when there is temperature transitions)

1. Ductile fractures show a rough dirty gray surface. This is due to how the crack propagates and creating very sharp ups and downs. This gives a dull feature due to the tips and valleys.
2. Plasticly deformed part that changed dimensions. This MAY happen when parts are subjected to static loadings with very slow strain rate (velocity of deformation).

e) **Micro-Fracture Features (Scanning Electron Microscope)**

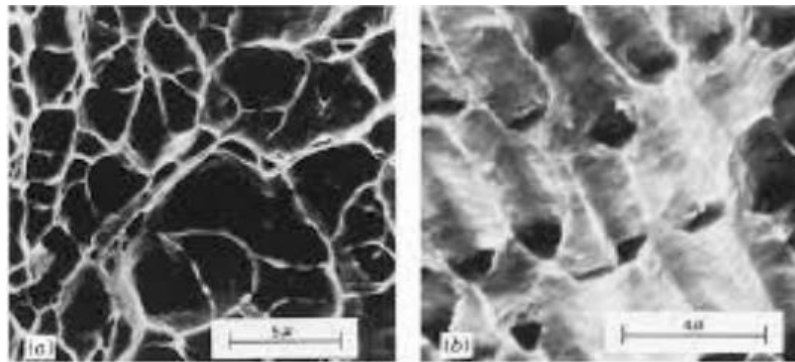
Under the high magnifications, the dark regions represent deep areas and bright regions represent protruding high areas.

Uniaxial tensile deformation of ductile materials:

1. Ductile fracture
2. Cleavage
3. Intergranular fracture

Mechanism:

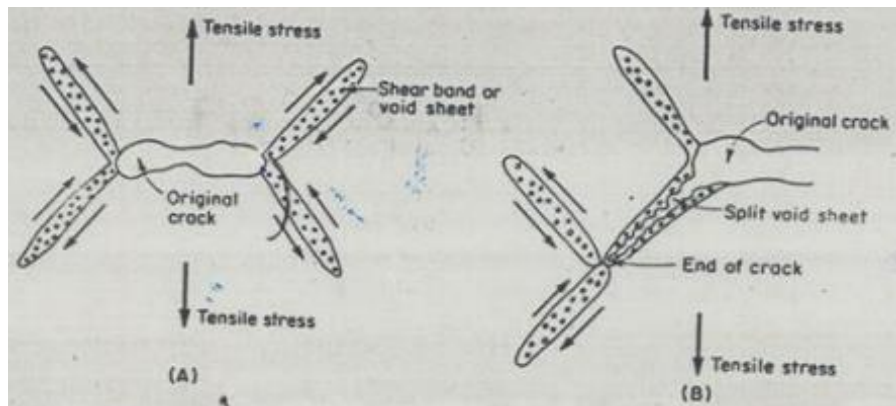
1. Inclusions in a ductile matrix
2. Void nucleation
3. Void growth
4. Strain localization between voids
5. Necking between voids
6. Void coalescence and fracture



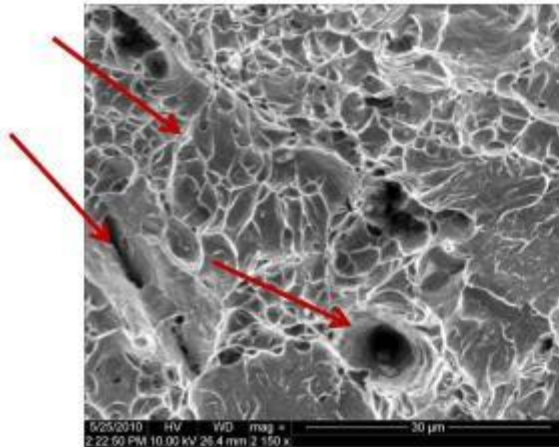
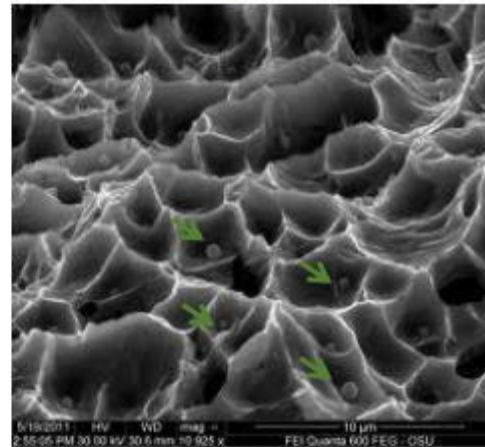
- a) Scanning electron fractography shows spherical dimples characteristic of ductile fracture resulting from uniaxial loads.
- b) Scanning electron fractography shows parabolic-shaped dimples characteristic of ductile fracture resulting from shear loading.

Void Sheet Mechanism in Ductile Fracture:

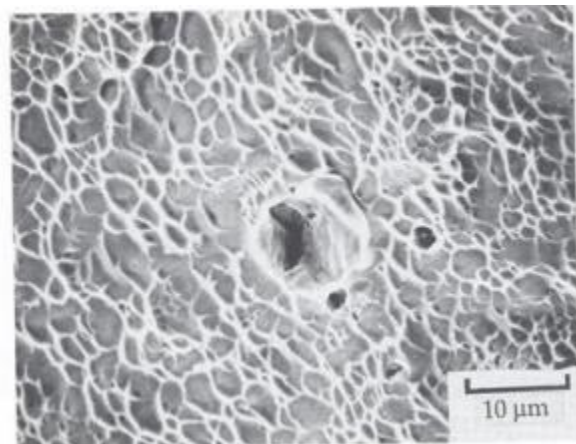
1. Necking
2. Shear Bands
3. Formation of Micro voids
4. Void Coalescence forming crack.
5. Propagation follows the same procedure.
6. Failure



- a) The stress concentration at the ends of a small crack initiated by void coalescence induces bands of shear strain at its ends. The highly concentrated strain in these bands nucleates pores inside the bands.
- b) The crack advances when a void sheet splits.

Dimples, Coarse Dimples and cracks**Dimples initiated at hard 1.0 μm particles**

c)



High magnification fractography of the shear region of a cup and cone fracture in austenitic steel.

f) Dimples Micro-Fracture Feature

Dimples used to be voids which opened to the neighboring ones. This feature under the micron level can be described as dimples that lead to trans granular ductile failure. Particles are also present at the center of the dimples, which are the favorable regions for void formations. Coarse dimples and finer dimples happen due to imbalance in the load distribution, cluster of precipitations or phases, or inclusions. Deep dimples represent a more ductile material compared to shallow dimples. Couplets are stretched (sheared) dimples that happen if the stresses have a dominant shear component over the axial component. When the tensile loads are higher in one side of the part, teared dimples occur. Pure tensile loadings will produce equiaxed dimples.

[When a ductile material fractures, it undergoes a process known as ductile fracture, which involves the propagation of micro voids and the coalescence of these voids into a crack. Dimples are the result of this process and are often observed on the fractured surface of a ductile material. The formation of dimples occurs because of the local stress concentration around the micro voids. As the stress around a void increase, the material near the void deforms more and eventually reaches its breaking point, resulting in the formation of a small crack. The crack then propagates along the material until it encounters another micro void, at which point it may branch or coalesce with the adjacent void. Dimples are typically round or elliptical in shape and can vary in size depending on the material and the conditions under which it fractured. The presence of dimples on the fracture surface of a ductile material is often taken as an indication that the material underwent ductile fracture.

Deep dimples on the fracture surface of a ductile material indicate a higher degree of plastic deformation before fracture compared to shallow dimples. Deep dimples are formed when the material around the micro void undergoes significant plastic deformation before the void coalesces into a crack. This indicates that the material has undergone a high degree of ductility before failure, which is desirable in many engineering applications. Shallow dimples are formed when the material has undergone less plastic deformation before fracture. This may indicate that the material was subjected to a higher rate of loading or that it had a lower degree of ductility, which can be undesirable in certain applications.

Couplets are a type of feature observed in ductile micro fractures that are formed due to the deformation and stretching of the material in the vicinity of the fracture. In a ductile micro

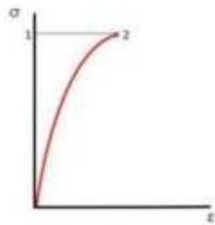
fracture, the material undergoes deformation and stretching near the site of the fracture, resulting in the formation of small, elongated cavities called "voids." These voids are often found in pairs, with one void on either side of the fracture surface, hence the name "couplet." Couplets are formed as a result of the stretching and thinning of the material near the fracture site, which causes the voids to elongate in the direction of the tensile stress.]

Brittle Material

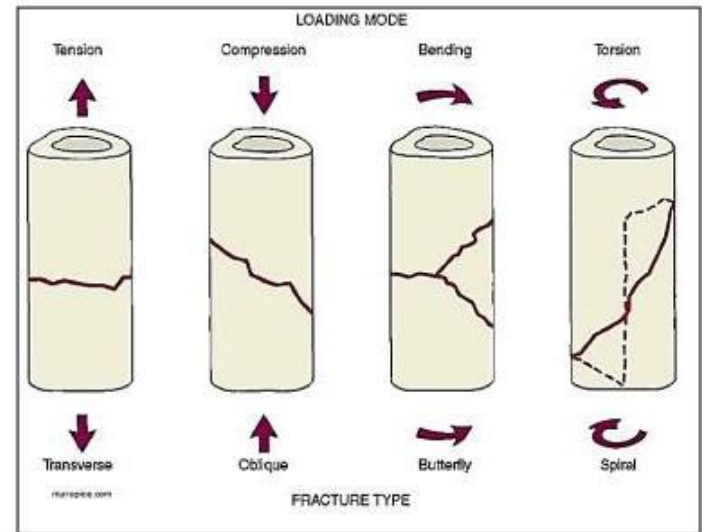
Crack propagation without plastic deformation. This means that there will be no shear zone formed. Plastic deformation requires energy as it is based on motion of dislocations. Therefore, crack propagating in brittle materials requires no energy.

a) Loading Modes

- Brittle fracture takes place without any appreciable deformation, & by rapid crack propagation. The direction of crack motion is very nearly perpendicular to the direction of the applied tensile stress & yields a relatively flat fracture surface
- When gradual tensile load is applied on material in tensile test, at the end of elastic limit, without any prior indication material breaks. This type of fracture is called as Brittle Fracture



(b) Brittle fracture in a mild steel.



b) Macro-Fracture Feature

Brittle fracture is the term given to static failures caused by crack propagation. A brittle fracture occurs when a pre-existing crack propagates very rapidly through the material. Such fractures release a lot of energy and can be very loud, explosive, and dangerous, as fragments of the material may be thrown long distances. In many cases, especially if the micro mechanism of cracking is cleavage, a macroscopic examination of the fracture surface may show lines (called "chevron marks") which lead back to the failure origin. Brittle crack extension, also called "cleavage", occurs by the splitting apart of planes in the atomic lattice. Each grain cleaves like a gemstone, giving a perfectly flat surface.

1. Shiny Facets (Line of atoms that were separated from adjacent planes) / Undeformed Flat Surfaces

In tensile stresses or loadings, the fracture will occur perpendicular to the axis of loading which is on planes of maximum tension where there is the maximum normal loads and no shear components (as resolves stresses) → This acts as separation not sliding.

EX/ Consider a Crystal (cube made out of planes of atoms) where there are stresses that exceed the pulling effect of two parallel planes, in this case the bonds will fracture by separation (zero sliding effect) and this occurs on planes of maximum tension.

EX/ Ionic Materials (Brittle): Ceramics are made out of cations and anions. Applying resolved shear stresses, atoms tend to move; however, when sliding motion bring a anion in front of another anion, this cannot be tolerated by the ionic material so repulsion occurs. This creates separation by breaking bonds perpendicularly in planes of atoms.

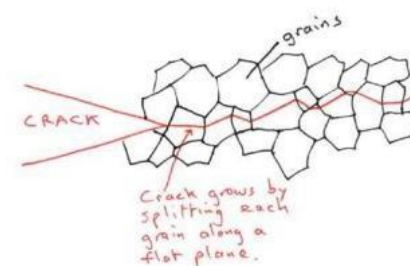
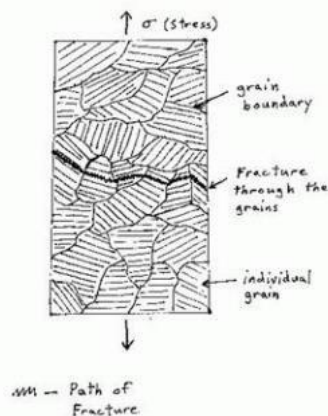
EX/ Metallic Material (Ductile): Metallic materials are mad out of atoms with the same electron configuration (there are no cation and anions). When resolved shear stresses are applied, planes will slide relative to each other that is accommodated by the sharing of electron clouds (no repulsion and equal attraction among atoms)

2. Chevron Marks

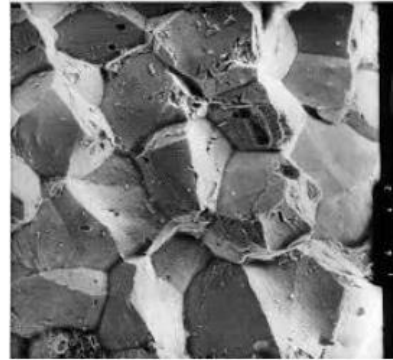
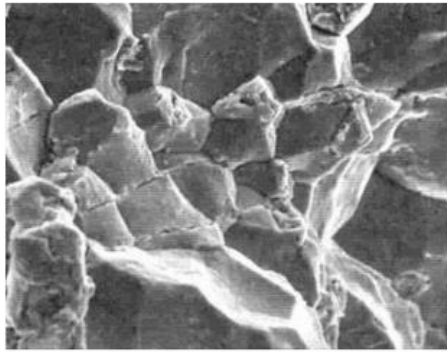
They are macro crack propagation path that is an indication of a brittle failure. There marks represent propagation of multiple cracks along different planes that are parallel (planes of maximum tension) to each other among different layers. Chevron marks point to the origin of the crack.

c) Types of Brittle Failures

Transgranular Fracture



Intergranular

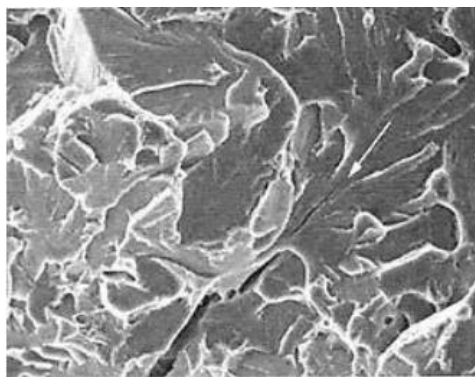


d) Micro-Fracture Features

In brittle material, cracks propagate trans granularly. Shortest path that requires minimum energy.

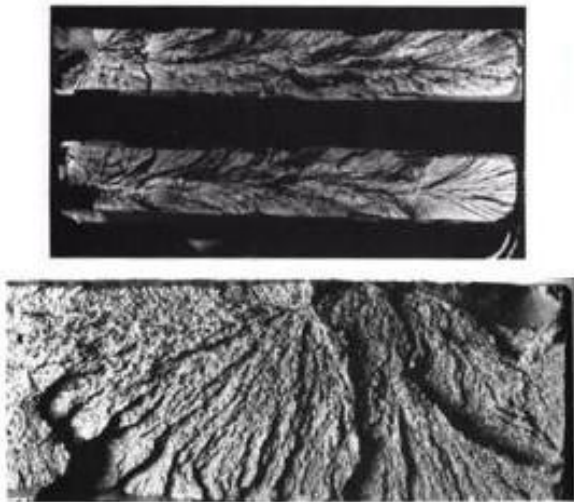
1. Cleavage Planes (Planes of maximum tension)
2. River Patterns (Bright edges with grey flat planes that represent different planes of micro crack propagations)

Inter granular brittle failure shows smooth boundaries where the crack propagated using brittle crack propagation along the grain boundaries, this is a problem. A defect (created a brittle phase) in the grain boundaries caused the crack to propagate in a brittle manner along those boundaries.

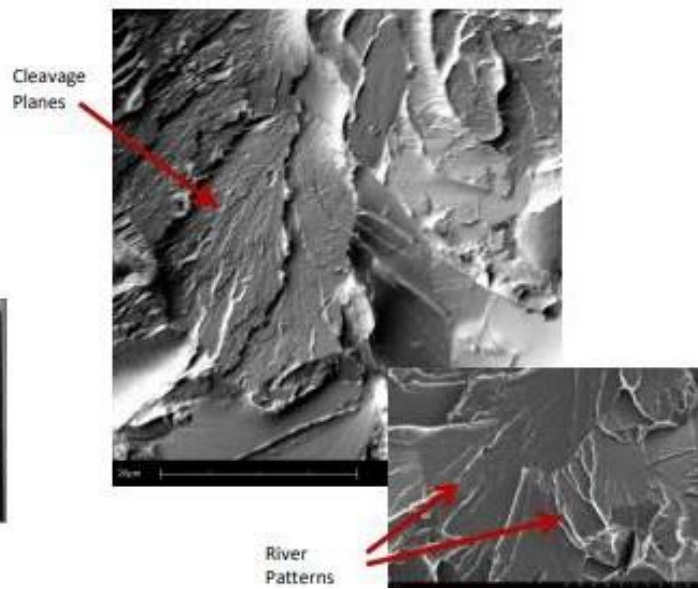


Cleavage Planes & River patterns

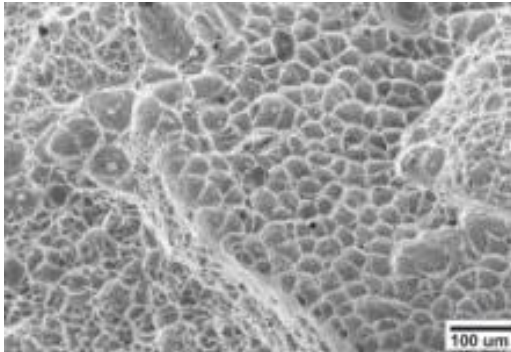
Brittle Macrofracture Feature Chevron Marks



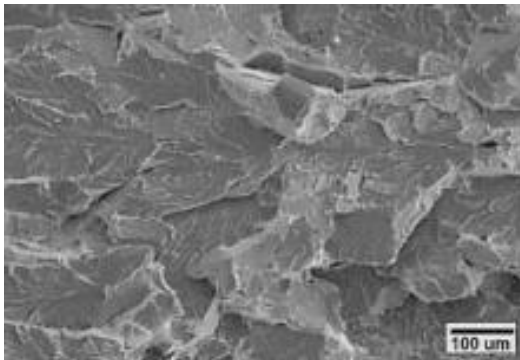
Brittle Microfracture features



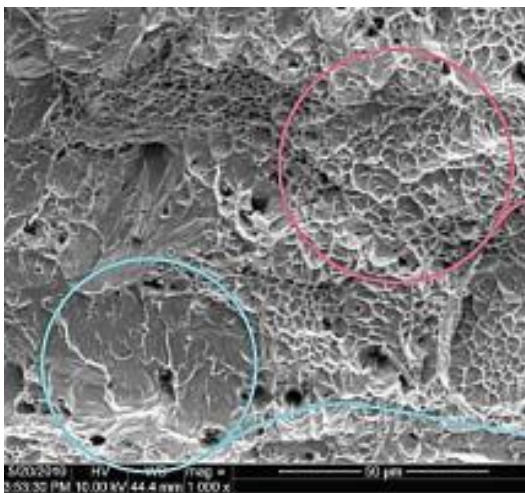
Test Your Understanding



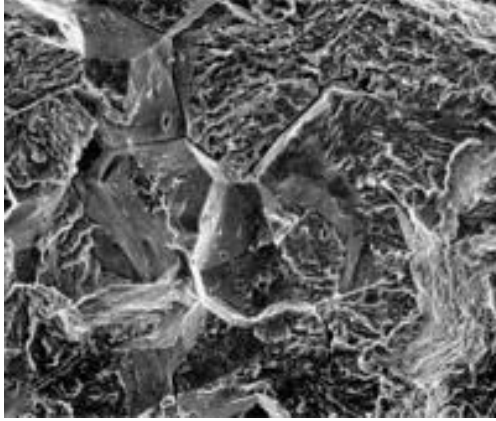
This is a micro-fracture feature for a ductile material failure as dimples are present. Those fine dimples have precipitates at their core. The high distribution of those particles facilitated the formation of millions of dimples which didn't have enough space to grow before they coalesce; therefore, they coalesce over a short distance.



This is a micro-fracture feature for a brittle material failure as it shows very flat cleavage planes and dull river patterns.



This is a combination of a ductile and brittle micro-fracture features. This indicates transitional behavior.



This is a micro-fracture feature of an inter granular ductile failure. This shows sheared planes of a ductile failure with couplets (sheared dimples).

Crack Formation Mechanisms

1. Cracks are the source of stress concentration. Cracks can be inherent/pre-existing (due to manufacturing processes). When there are no pre-existing cracks, micro-crack-like formation can also take place in further treatment steps. Consider strain hardening where there are movement and generation of new dislocations (minimum dislocation density in any material is 10^5 per cm^2), this will create fluctuations of full and half planes (dislocation line is a 1D defect: plane that separates between half plane and full plane is called dislocation line). When dislocation lines (half planes) are adjacent, 3D defects will occur in the form of micro cracks. This happens when we reach the sigma ultimate of the material.

2. Zener's Mechanism:

Some metallic materials cannot slip, such as CPH and HCP, due to the slip planes not touching. Such materials deform by twinning mechanism. The perpendicular boundary the formed as a result of the twinning effect of these planes is called twinning boundary. Dislocations that are moving on a slip plane facing a twined region will stop. These dislocations will pile up at the contact point between the twin boundary until the boundary cannot tolerate the applied stresses, so it cracks.

3. Cottrell's Mechanism

Different dislocations moving on various slip plane will come to a stop when they face a non-slip plane. Dislocations will pile up and create internally hairy cracks.

Summary:

Crack Propagation Mechanism:

Ductile → Void Sheet Mechanism [**Macro:** Rough Dirty Grey Surface; **Micro:** Dimples or Couplets]

Brittle → Crack Cleavage Propagation [**Macro:** Shiny Facets and Chevron Marks; **Micro:** Cleavage Planes and River Patterns]

A big difference between brittle fracture and ductile fracture is that in brittle fracture there is no macroscopic deformation. So, you can take the broken parts and put them back together again, like making a jigsaw. You can't do this after a ductile fracture because the shape of the pieces has changed.

Self-Healing Material Techniques (Foam Products)

1. In monolithic foam ceramics, manually adding circular shapes for when the cracks propagate and hits the empty spheres will relieve the stress concentration below yielding; therefore, adding toughness and increasing σ yield to the whole system. This can be achieved by a designed infiltrating process where cobalt or polymers infiltrate within or by mixing the powders (after mixing, we center. In the case of cobalt, it melts → encapsulates ceramic particles → creates a filling that stops cracks propagation). Polymers are thermoplastic material that can be added to ceramics to stop crack propagation by absorbing the energy. Polymers added to ceramics will not carry the applied loads (limitation: cannot be used at high temperatures as the melting point of polymers is very low, 24 °C).
2. Widening rho (ρ) and making it circular-shaped will reduce the stress concentrations to minimum at the crack tip. Drilling designed holes at the crack tips creates the widest possible rho (ρ). This absorbs all the energy that was at the crack tip.

Example: Gears sometimes suffer from cracks. Instead of replacing the part, a very tiny whole can be drilled at the crack tip (only if the crack propagation direction is not harmful). This stops the crack from propagating.

Example: in the early days, small, cracked car windows were repaired by drilling small holes at the end of the crack. This stops the crack from propagating.